HARD

GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY

What is the diameter of the circle in the *xy*-plane with equation $x - 5^2 + y - 3^2 = 16$?

A. 4

B. 8

C. 16

D. 32

Q2 GEOMETRY AND TRIGONOMETRY

In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

A. $-\frac{\sqrt{3}}{3}$

B. $-\frac{3}{\sqrt{3}}$

C. $\frac{\sqrt{3}}{3}$

D. $\frac{3}{\sqrt{3}}$

Q3 GEOMETRY AND TRIGONOMETRY

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $7x$ - 250°. The sum of the measures of four of the eight angles is k°. Which of the following could NOT be equivalent to k, for all values of x?

A. $-14x + 1{,}540$

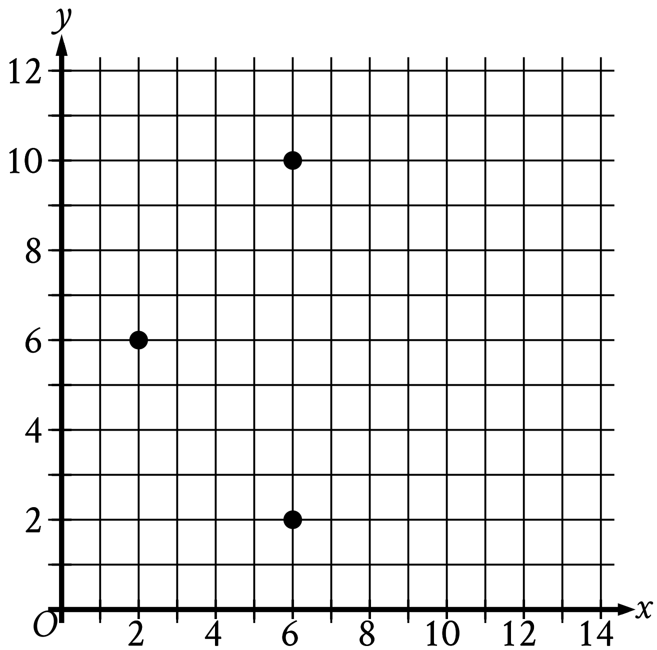
B. $14x - 320$

C. $-28x + 1{,}720$

D. 360

Q4

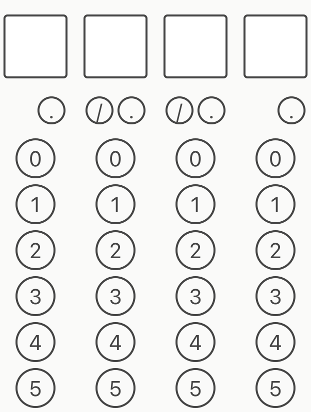
GRID-IN GEOMETRY AND TRIGONOMETRY



- From left to right, the 3 points have the following coordinates:
  - (2 comma 6)
  - (6 comma 10)
  - (6 comma 2)

The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE



6 6 6 6
7 7 7 7
8 8 8 8
9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 GEOMETRY AND TRIGONOMETRY

In the xy-plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

A. 5

B. 6.5

C. $\sqrt{40}$

D. $\sqrt{50}$

Q6 GEOMETRY AND TRIGONOMETRY

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

A. 9

B. $9\sqrt{2}$

C. 18

D. $18\sqrt{2}$

Q7

GRID-IN GEOMETRY AND TRIGONOMETRY

In right triangle ABC, angle C is the right angle and $BC = 162$. Point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC and $CE = 2AE$. What is the length of line segment DE?

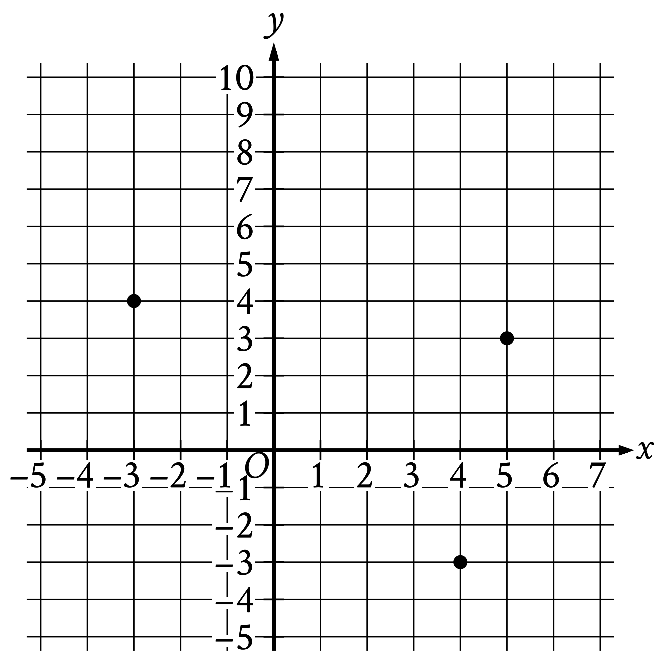
STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY



- Moving from left to right, the points have the following coordinates:
  - (negative 3 comma 4)
  - (4 comma negative 3)
  - (5 comma 3)

What is the area, in square units, of the triangle formed by connecting the three points shown?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN GEOMETRY AND TRIGONOMETRY

Circle A in the *xy*-plane has the equation $x + 5^2 + y - 5^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the *xy*-plane is $x + 5^2 + y - 5^2 = k$, where k is a constant. What is the value of k?

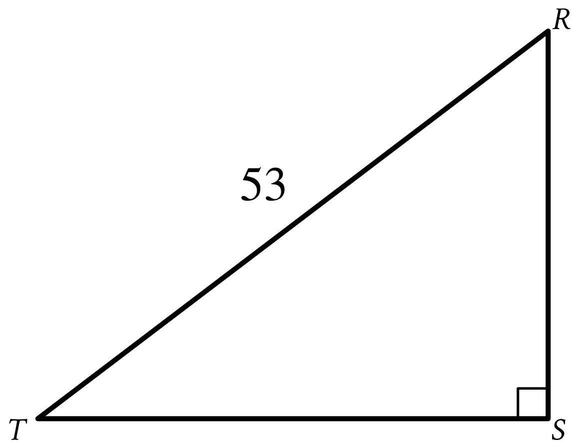
STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN GEOMETRY AND TRIGONOMETRY

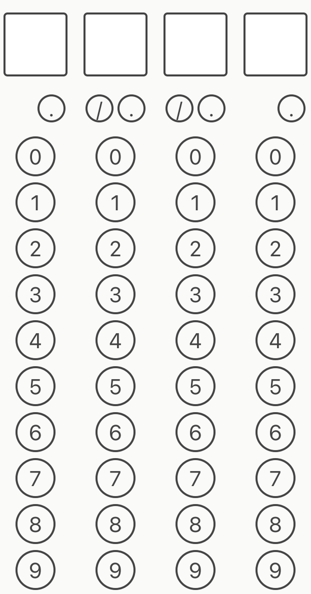


Note: Figure not drawn to scale.

- Angle upper S is a right angle.
- The length of side upper R upper T is 53.
- A note indicates the figure is not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.